EVERY TWO-COLOURING OF THE PLANE CONTAINS A MONOCHROMATIC DEGENERATE TRIPLE WITH SIDES 1, 3 AND 4

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Abstract. We prove that every two-colouring of the Euclidean plane contains three collinear points of one colour with consecutive distances 1 and 3. The argument works on the triangular lattice $\Lambda = \mathbb{Z}[\omega]$ together with a single rotated copy $q\Lambda$, where $q = (3+5\omega)/7$; the two lattices share a sublattice of index 7. We show that, in one of its three directions, every line of a unit triangular lattice is coloured in runs of exactly four, and that consecutive such lines are always related in the same way. Then any two points at distance 8 have the same colour, which forces a monochromatic unit triangle.

1. Introduction

For $a, b, c > 0$ write $T(a, b, c)$ for a configuration of three points with pairwise distances a, b, c, the degenerate (collinear) case allowed, and $T(s) = T(s, s, s)$. A colouring *contains* a configuration if some congruent copy is monochromatic. Erdős, Graham, Montgomery, Rothschild, Spencer and Straus [1] initiated the study of which configurations every two-colouring of the plane must contain. Currier, Moore and Yip [2] proved that every two-colouring of the plane contains a monochromatic three-term arithmetic progression, that is, a monochromatic $T(1, 1, 2)$, which was an inspiration to this research.

Theorem 1.1. *Every two-colouring of* $\mathbb{R}^2$ *contains a monochromatic* $T(1, 3, 4)$.

Scaling gives a monochromatic $T(a, 3a, 4a)$ for every $a > 0$. The proof has four steps.

(1) *Rule A* (Section 3): along the long direction of any unit triangular lattice, monochromatic runs have length exactly four.
(2) *Rows* (Section 4): runs of four alternate along the long direction, and each pair of consecutive rows is related by a colour flip along one of the two other directions.
(3) *Rule B* (Section 5): this flip direction is the same for all pairs of consecutive rows; equivalently, every open pair (defined in Section 5) has two colours.
(4) *Distance eight* (Section 6): a flip repeated eight times along one direction returns the colour, so any two points at distance 8 have the same colour, and every unit equilateral triangle is monochromatic.

Theorem 1.1 is proved at the end of Section 6; Section 3.2, the second proof of Rule A, and Section 7, on verification, are not needed for it. The structural input is the description of $T(1, 3, 4)$-avoiding colourings of the triangular lattice from the author's thesis [3], and the idea of overlapping a lattice with a rotated copy proposed in its conclusion. Both finite arguments use only the base lattice and one rotation.

2. Preliminaries

Let $\omega = e^{i\pi/3}$, so $\omega^2 = \omega - 1$ and $\bar{\omega} = 1 - \omega$, and $\Lambda = \mathbb{Z} + \mathbb{Z}\omega$ with $|a + b\omega|^2 = a^2 + ab + b^2$. A *unit triangular lattice* is a set $z_0 + u\Lambda$ with $|u| = 1$; its three *directions* are $u, u\omega, u\omega^2$, and its *rows* in a direction are the lines parallel to it.

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The rotation. Since $\bar{\omega} = 1 - \omega$, the conjugate of $2 + \omega$ is $3 - \omega$; both have absolute value $\sqrt{7}$, as $|2 + \omega|^2 = 4 + 2 + 1 = 7$. Put

$$q = \frac{2+\omega}{3-\omega} = \frac{(2+\omega)^2}{7} = \frac{3+5\omega}{7}, \qquad |q| = 1.$$

Since $2 + \omega = q(3 - \omega)$, the sublattice $(2 + \omega)\Lambda$ of index 7 lies in both Λ and $q\Lambda$ (Figure 1). Every point of $\Lambda \cup q\Lambda$ is $[a, b] := (a + b\omega)/7$ with $a, b \in \mathbb{Z}$; it lies in Λ iff $7 \mid a, b$, and in $q\Lambda$ iff $49 \mid 8a + 5b$ and $49 \mid 3b - 5a$. Points of $q\Lambda$ are written $q(m + n\omega)$.

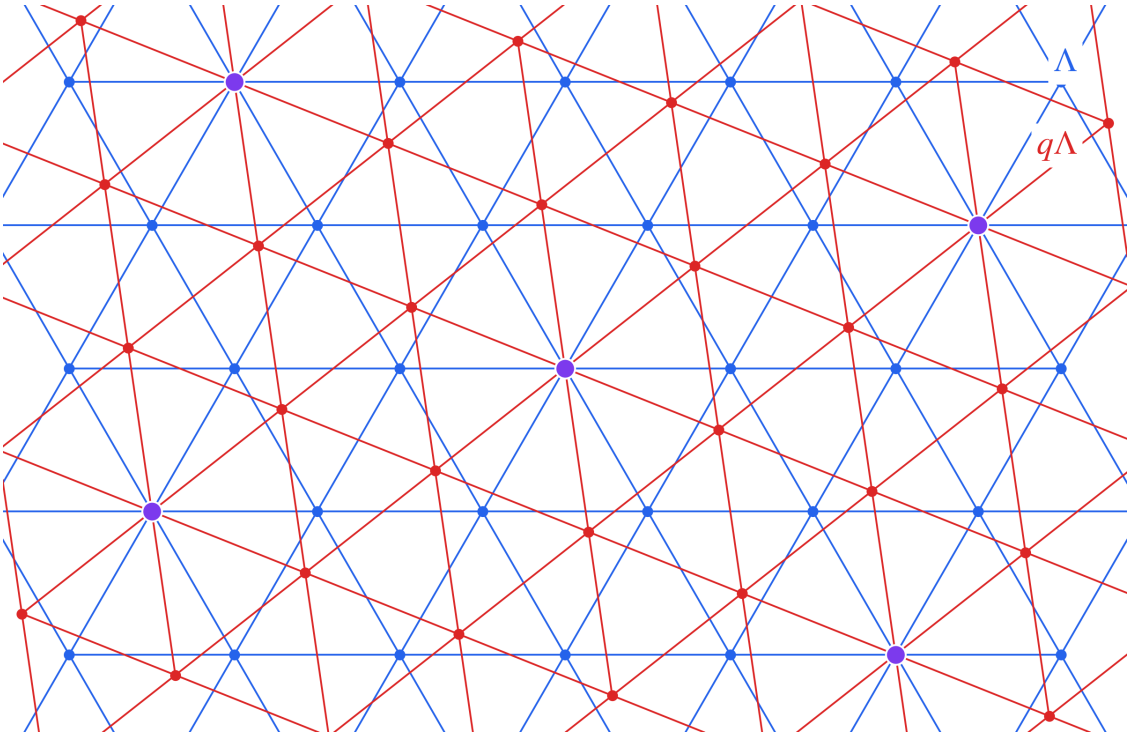


FIGURE 1. Λ (blue) and $q\Lambda$ (red). The purple points form the common sublattice $(2 + \omega)\Lambda = q(3 - \omega)\Lambda$, of index 7 in both.

Forbidden triangles. A two-colouring of the plane is *admissible* if it contains no monochromatic $T(1, 3, 4)$; we assume one is given and derive a contradiction. The following is the eight-point argument of [1].

Lemma 2.1 (bridge)**.** *An admissible colouring contains no monochromatic $T(1)$, $T(3)$ or $T(4)$.*

Proof. With $c = 1$, $b = 4$ put $A = 0$, $B = c$, $C = b$, $D = c - c\omega$, $E = b + (c - b)\omega$, $F = b - b\omega$, $G = c - b\omega$, $H = b + c - b\omega$. The triples $ABC, ADF, BDG, CEF, FGH, BEH$ are copies of $T(1, 3, 4)$, and $ABD, ACF, BCE, DFG, EFH, BGH$ are equilateral of sides $1, 4, 3, 3, 1, 4$; in particular ABD, BCE, ACF have sides 1, 3, 4. Assume there is no monochromatic $T(1, 3, 4)$, so each of the six collinear triples contains both colours. If A, B, D are orange, then ABC, ADF, BDG give C, F, G yellow, then CEF, FGH give E, H orange, and BEH is orange. If A, C, F are orange, then ABC, ADF, CEF give B, D, E yellow, then BDG, BEH give G, H orange, and FGH is orange. If B, C, E are orange, then ABC, CEF, BEH give A, F, H yellow, then ADF gives D orange and BDG gives G yellow, and FGH is yellow. Every equilateral triangle of side 1, 3 or 4 is congruent to ABD, BCE or ACF, and exchanging colours covers yellow triangles. □

A *forbidden triangle* is a $T(1)$, $T(3)$, $T(4)$ or $T(1,3,4)$. Every finite argument below uses one rule:

> **(F)** If two vertices of a forbidden triangle have the same colour, the third has the other colour.

The structure theorem. We use the following result from the thesis.

Theorem 2.2 ([3, Theorem 3.13])**.** *In a $T(1,3,4)$-avoiding two-colouring of a unit triangular lattice, one of the three directions, the* long direction*, has all maximal monochromatic runs of length 3 or 4; in each of the other two directions every run has length 1 or 2.*

The restriction of an admissible colouring to any unit triangular lattice is $T(1,3,4)$-avoiding, so Theorem 2.2 applies to Λ, to $q\Lambda$, and to all their images under isometries.

3. Rule A

Theorem 3.1 (Rule A)**.** *In an admissible colouring, no line of a unit triangular lattice carries a maximal monochromatic run of exactly three points.*

Lemma 3.2 (the row)**.** *If such a run exists then, after an isometry and possibly exchanging colours, $0,1,2$ are orange and $-3,-2,-1$ and $3,4,5$ are yellow.*

Proof. Normalise the run to $0,1,2$ orange with $-1,3$ yellow, in a lattice Λ. A run of length three lies in the long direction by Theorem 2.2, and runs there have length at least three, so the yellow runs through -1 and 3 contain $-3,-2,-1$ and $3,4,5$. □

Lemma 3.3 (the boundary)**.** *In this situation $1-\omega, 2-\omega, \omega, 1+\omega$ are yellow, at least one of $-\omega, 3-\omega$ is yellow, and at least one of $-1+\omega, 2+\omega$ is yellow.*

Proof. The unit triangles $\{0,1,1-\omega\}$, $\{1,2,2-\omega\}$, $\{0,1,\omega\}$, $\{1,2,1+\omega\}$ have two orange vertices. The rows $\mathbb{Z}\pm\omega$ are lines of the long direction, so the yellow runs through $1-\omega, 2-\omega$ and through $\omega, 1+\omega$ have length at least three. □

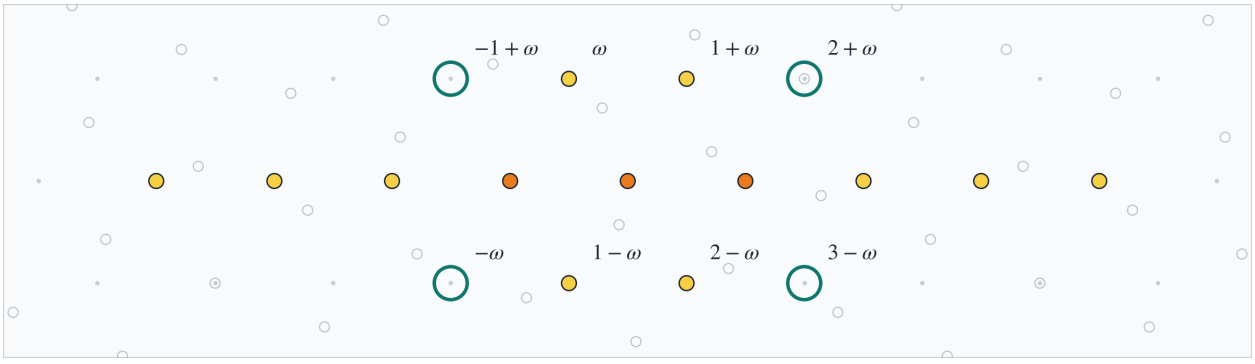


FIGURE 2. Rule A: the row $-3,\dots,5$, the four points forced yellow by unit triangles, and the four boundary points (ringed), of which each pair below and above contains a yellow point.

Lemma 3.4 (symmetries)**.** *The reflections $\tau(z)=\bar{z}$ and $\sigma(z)=2-\bar{z}$ preserve the row of Lemma 3.2 with its colours. σ swaps $-\omega \leftrightarrow 3-\omega$ and $-1+\omega \leftrightarrow 2+\omega$; τ swaps $-\omega \leftrightarrow -1+\omega$ and $3-\omega \leftrightarrow 2+\omega$. If χ is admissible, so is $\chi\circ\rho$ for every isometry ρ.*

Proof. Both maps preserve the real axis, τ pointwise and σ as $x \mapsto 2-x$, which preserves the colours of $-3,\dots,5$. For example $\sigma(-\omega) = 2-\overline{-\omega} = 2-(\omega-1) = 3-\omega$. □

Proof of Theorem 3.1. By Lemma 3.3 the colours of $(-\omega,\ 3-\omega,\ -1+\omega,\ 2+\omega)$ have a yellow point in each pair: nine possibilities. The group $\{1, \sigma, \tau, \sigma\tau\}$ permutes them in four orbits, of sizes $1, 4, 2, 2$, with representatives: all four yellow; $3-\omega, -1+\omega, 2+\omega$ yellow; $3-\omega, 2+\omega$ yellow; $-\omega, 2+\omega$ yellow. Replacing χ by $\chi \circ \rho$ we may assume the boundary is one of these. Each is refuted inside $\Lambda \cup q\Lambda$ by rule (F), with one case split; the complete deductions are in Section 3.1. □

3.1. **The four cases.** Points are named in Λ or as $q(m+n\omega)$. The row $-3, \dots, 5$ is as in Lemma 3.2. In each case, deductions are listed before the case split and in each subcase; each subcase ends in a contradiction.

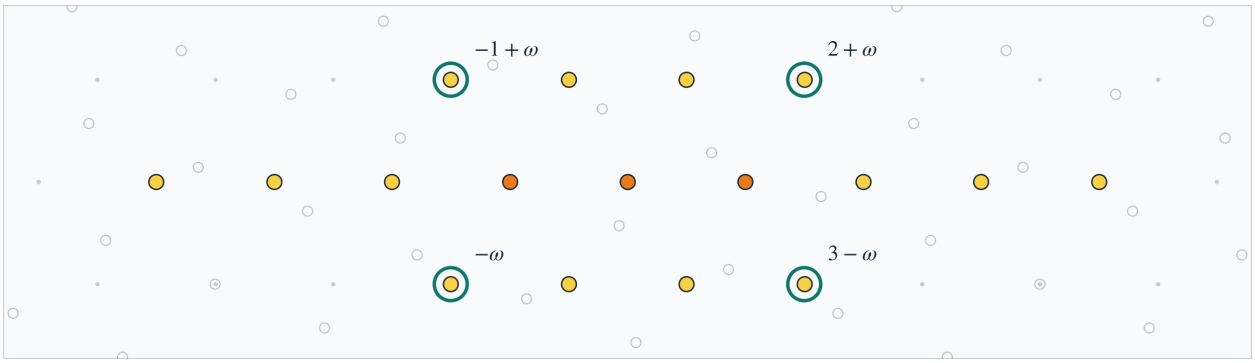


FIGURE 3. Rule A, case 1: the row, the four forced points, and the boundary colouring of this case (ringed).

Case 1: yellow $-\omega$, $3-\omega$, $-1+\omega$, $2+\omega$.

	point	colour	because
1	$-1-\omega$	orange	$T(1)$ with -2, -1 (yellow)
2	ω	yellow	$T(1)$ with 0, 1 (orange)
3	$1+\omega$	yellow	$T(1)$ with 1, 2 (orange)
4	$2-\omega$	yellow	$T(1)$ with 1, 2 (orange)
5	7	orange	$T(1,3,4)$ with 3, 4 (yellow)
6	2ω	orange	$T(3)$ with $-\omega$, $3-\omega$ (yellow)
7	$3-2\omega$	orange	$T(1)$ with $2-\omega$, $3-\omega$ (yellow)
8	$-1+2\omega$	orange	$T(1)$ with $-1+\omega$, ω (yellow)
9	$1+2\omega$	orange	$T(1)$ with $1+\omega$, $2+\omega$ (yellow)
10	$-1+3\omega$	yellow	$T(1,3,4)$ with $-1-\omega$, $-1+2\omega$ (orange)
11	$4+2\omega$	yellow	$T(1,3,4)$ with 2ω, $1+2\omega$ (orange)

Subcase: $q(2+\omega)$ yellow.

	point	colour	because
1	$q(5-2\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(2+\omega)$ (yellow)
2	$q(4-\omega)$	yellow	$T(1,3,4)$ with $q(5-2\omega)$, 7 (orange)
3	$q(3\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (yellow)
4	$q(-\omega)$	orange	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (yellow)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all orange and form $T(1,3,4)$.

Subcase: $q(2+\omega)$ orange.

	point	colour	because
1	$q(2)$	yellow	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (orange)
2	$q(2-2\omega)$	yellow	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (orange)
3	$q(3)$	orange	$T(1)$ with $q(2)$, $2+\omega$ (yellow)
4	$q(5-2\omega)$	orange	$T(1,3,4)$ with $q(2-2\omega)$, $4+2\omega$ (yellow)
5	$q(3\omega)$	yellow	$T(3)$ with 0, $q(3)$ (orange)
6	$q(4-\omega)$	yellow	$T(1,3,4)$ with $q(5-2\omega)$, 7 (orange)

Contradiction: $q(3\omega)$, $-1+3\omega$, $q(4-\omega)$ are all yellow and form $T(1,3,4)$.

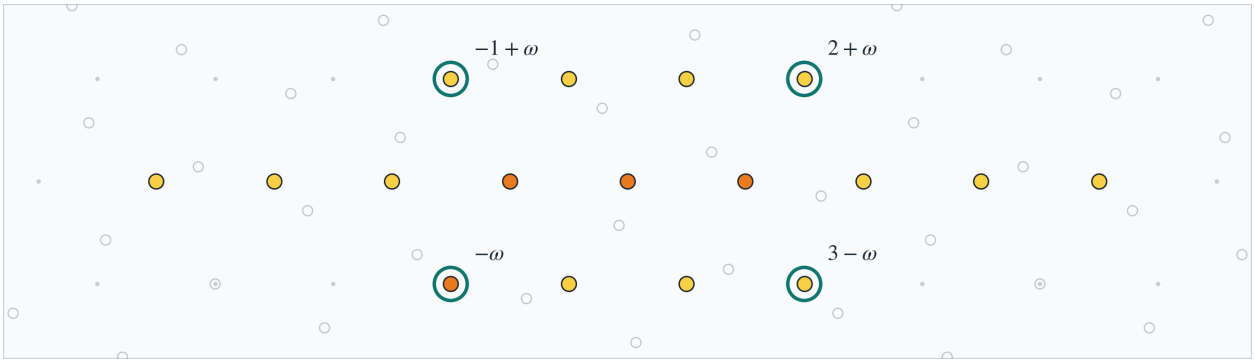


FIGURE 4. Rule A, case 2: the row, the four forced points, and the boundary colouring of this case (ringed).

Case 2: yellow $3-\omega$, $-1+\omega$, $2+\omega$*; orange* $-\omega$.

	point	colour	because
1	$-1-\omega$	orange	$T(1)$ with -2, -1 (yellow)
2	ω	yellow	$T(1)$ with 0, 1 (orange)
3	$1+\omega$	yellow	$T(1)$ with 1, 2 (orange)
4	$2-\omega$	yellow	$T(1)$ with 1, 2 (orange)
5	7	orange	$T(1,3,4)$ with 3, 4 (yellow)
6	$3-2\omega$	orange	$T(1)$ with $2-\omega$, $3-\omega$ (yellow)
7	$-1+2\omega$	orange	$T(1)$ with $-1+\omega$, ω (yellow)
8	$1+2\omega$	orange	$T(1)$ with $1+\omega$, $2+\omega$ (yellow)
9	$-1+3\omega$	yellow	$T(1,3,4)$ with $-1-\omega$, $-1+2\omega$ (orange)
10	2ω	orange	$T(1)$ with ω, $1+\omega$ (yellow)
11	$4+2\omega$	yellow	$T(1,3,4)$ with 2ω, $1+2\omega$ (orange)

Subcase: $q(2+\omega)$ yellow.

	point	colour	because
1	$q(5-2\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(2+\omega)$ (yellow)
2	$q(4-\omega)$	yellow	$T(1,3,4)$ with $q(5-2\omega)$, 7 (orange)
3	$q(3\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (yellow)
4	$q(-\omega)$	orange	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (yellow)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all orange and form $T(1,3,4)$.

Subcase: $q(2+\omega)$ orange.

	point	colour	because
1	$q(2)$	yellow	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (orange)
2	$q(2-2\omega)$	yellow	$T(1,3,4)$ with $q(2+\omega)$, $3-2\omega$ (orange)
3	$q(3)$	orange	$T(1)$ with $q(2)$, $2+\omega$ (yellow)
4	$q(5-2\omega)$	orange	$T(1,3,4)$ with $q(2-2\omega)$, $4+2\omega$ (yellow)
5	$q(3\omega)$	yellow	$T(3)$ with 0, $q(3)$ (orange)
6	$q(4-\omega)$	yellow	$T(1,3,4)$ with $q(5-2\omega)$, 7 (orange)

Contradiction: $q(3\omega)$, $-1+3\omega$, $q(4-\omega)$ are all yellow and form $T(1,3,4)$.

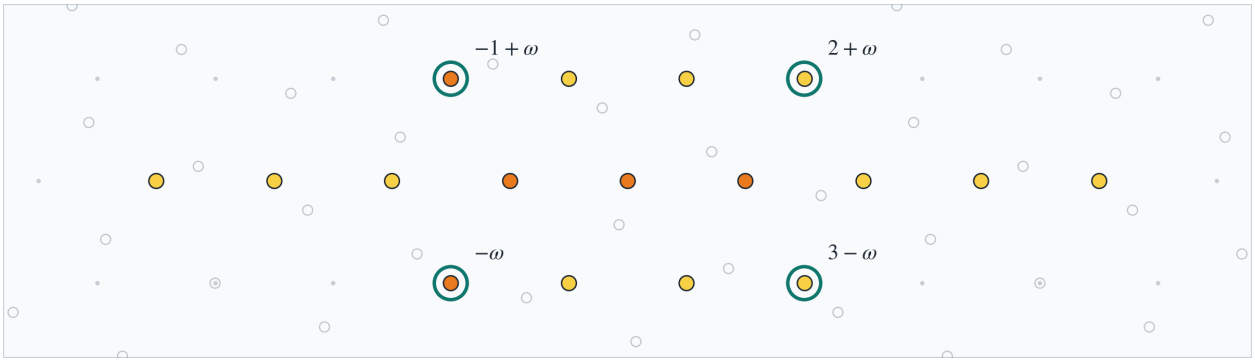


FIGURE 5. Rule A, case 3: the row, the four forced points, and the boundary colouring of this case (ringed).

Case 3: yellow $3-\omega$, $2+\omega$*; orange* $-\omega$, $-1+\omega$.

	point	colour	because
1	$3-4\omega$	orange	$T(4)$ with -1, 3 (yellow)
2	ω	yellow	$T(1)$ with $-1+\omega$, 0 (orange)
3	$1+\omega$	yellow	$T(1)$ with 1, 2 (orange)
4	7	orange	$T(1,3,4)$ with 3, 4 (yellow)
5	$5-\omega$	orange	$T(1)$ with 4, 5 (yellow)
6	$4-5\omega$	yellow	$T(1,3,4)$ with $-\omega$, $3-4\omega$ (orange)
7	$1+2\omega$	orange	$T(1)$ with $1+\omega$, $2+\omega$ (yellow)
8	2ω	orange	$T(1)$ with ω, $1+\omega$ (yellow)
9	$4+2\omega$	yellow	$T(1,3,4)$ with 2ω, $1+2\omega$ (orange)

Subcase: $q(5-5\omega)$ yellow.

	point	colour	because
1	$q(1-\omega)$	orange	$T(4)$ with $4-5\omega$, $q(5-5\omega)$ (yellow)
2	$q(4-5\omega)$	orange	$T(1,3,4)$ with $4-5\omega$, $q(5-5\omega)$ (yellow)
3	$q(-\omega)$	yellow	$T(1)$ with 0, $q(1-\omega)$ (orange)
4	$q(4-\omega)$	yellow	$T(4)$ with $q(4-5\omega)$, 7 (orange)

Contradiction: $q(-\omega)$, $2+\omega$, $q(4-\omega)$ are all yellow and form $T(1,3,4)$.

Subcase: $q(5-5\omega)$ orange.

	point	colour	because
1	$q(5-\omega)$	yellow	$T(1,3,4)$ with $5-\omega$, $q(5-5\omega)$ (orange)
2	$q(5-2\omega)$	yellow	$T(3)$ with $q(5-5\omega)$, 7 (orange)

Contradiction: $q(5-\omega)$, $q(5-2\omega)$, $4+2\omega$ are all yellow and form $T(1)$.

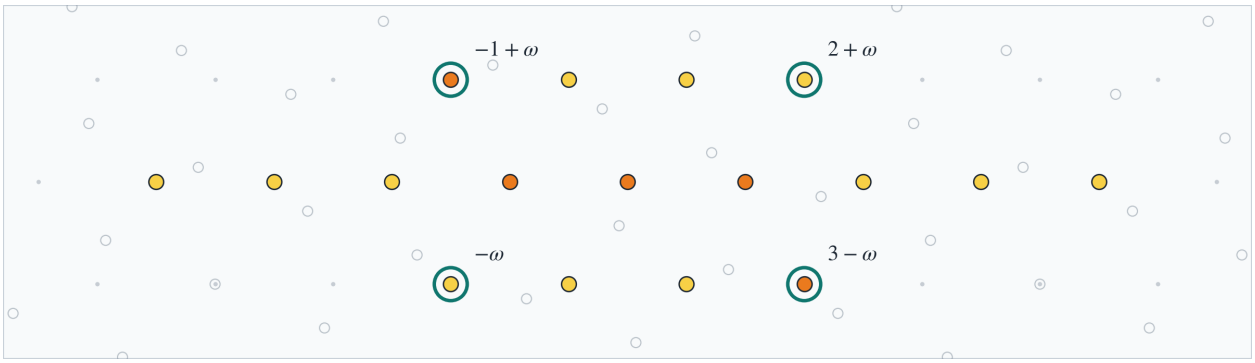


FIGURE 6. Rule A, case 4: the row, the four forced points, and the boundary colouring of this case (ringed).

Case 4: yellow $-\omega$, $2+\omega$*; orange* $3-\omega$, $-1+\omega$.

	point	colour	because
1	$3-4\omega$	orange	$T(1,3,4)$ with -1, $-\omega$ (yellow)
2	ω	yellow	$T(1)$ with $-1+\omega$, 0 (orange)
3	$3-3\omega$	yellow	$T(1,3,4)$ with $-1+\omega$, 0 (orange)
4	$1+\omega$	yellow	$T(1)$ with 1, 2 (orange)
5	$-1+3\omega$	yellow	$T(1,3,4)$ with 2, $3-\omega$ (orange)
6	$6-4\omega$	yellow	$T(1,3,4)$ with 2, $3-\omega$ (orange)
7	7	orange	$T(1,3,4)$ with 3, 4 (yellow)
8	$5-\omega$	orange	$T(1)$ with 4, 5 (yellow)
9	$1+2\omega$	orange	$T(1)$ with $1+\omega$, $2+\omega$ (yellow)
10	2ω	orange	$T(1)$ with ω, $1+\omega$ (yellow)
11	$4+2\omega$	yellow	$T(1,3,4)$ with 2ω, $1+2\omega$ (orange)

Subcase: $q(4-\omega)$ yellow.

	point	colour	because
1	$q(3\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(4-\omega)$ (yellow)
2	$q(-\omega)$	orange	$T(1,3,4)$ with $2+\omega$, $q(4-\omega)$ (yellow)

Contradiction: $q(3\omega)$, 0, $q(-\omega)$ are all orange and form $T(1,3,4)$.

Subcase: $q(4-\omega)$ orange.

	point	colour	because
1	$q(5-2\omega)$	yellow	$T(1,3,4)$ with $q(4-\omega)$, 7 (orange)
2	$q(4-5\omega)$	yellow	$T(4)$ with $q(4-\omega)$, 7 (orange)
3	$q(2+\omega)$	orange	$T(1,3,4)$ with $-1+3\omega$, $q(5-2\omega)$ (yellow)
4	$q(1-2\omega)$	orange	$T(4)$ with $-1+3\omega$, $q(5-2\omega)$ (yellow)
5	$q(2-2\omega)$	orange	$T(1,3,4)$ with $q(5-2\omega)$, $4+2\omega$ (yellow)
6	$q(5-\omega)$	orange	$T(1)$ with $q(5-2\omega)$, $4+2\omega$ (yellow)
7	$q(6-3\omega)$	orange	$T(1)$ with $q(5-2\omega)$, $4+2\omega$ (yellow)
8	$q(3-5\omega)$	orange	$T(1)$ with $q(4-5\omega)$, $6-4\omega$ (yellow)
9	$q(3)$	yellow	$T(1,3,4)$ with $q(2+\omega)$, $q(6-3\omega)$ (orange)
10	$q(1-\omega)$	yellow	$T(1)$ with $q(1-2\omega)$, $q(2-2\omega)$ (orange)
11	$q(5-5\omega)$	yellow	$T(1,3,4)$ with $q(5-\omega)$, $5-\omega$ (orange)
12	$q(3-4\omega)$	orange	$T(1,3,4)$ with $q(3)$, $2+\omega$ (yellow)
13	$4-5\omega$	orange	$T(4)$ with $q(1-\omega)$, $q(5-5\omega)$ (yellow)
14	$7-7\omega$	yellow	$T(1,3,4)$ with $q(3-4\omega)$, $q(3-5\omega)$ (orange)
15	$4-4\omega$	yellow	$T(1)$ with $3-4\omega$, $4-5\omega$ (orange)

Contradiction: $3-3\omega$, $4-4\omega$, $7-7\omega$ are all yellow and form $T(1,3,4)$.

3.2. **A second proof in two splits.** A second proof uses only σ: assume $3-\omega$ is yellow, by σ and Lemma 3.3, then split on the point q. If q is orange, rule (F) alone gives a contradiction; if q is yellow, one more split finishes. It needs two splits in total but 81 points instead of 49. In particular q is yellow in every such configuration.

	point	colour	because
1	-6	orange	$T(1,3,4)$ with -3, -2 (yellow)
2	$-3+\omega$	orange	$T(1)$ with -3, -2 (yellow)
3	$-2-\omega$	orange	$T(1)$ with -3, -2 (yellow)
4	-5	orange	$T(1,3,4)$ with -2, -1 (yellow)
5	$-2+\omega$	orange	$T(1)$ with -2, -1 (yellow)
6	$-1-\omega$	orange	$T(1)$ with -2, -1 (yellow)
7	$-1+4\omega$	orange	$T(4)$ with -1, 3 (yellow)
8	ω	yellow	$T(1)$ with 0, 1 (orange)
9	$1+\omega$	yellow	$T(1)$ with 1, 2 (orange)
10	$2-\omega$	yellow	$T(1)$ with 1, 2 (orange)
11	$3-2\omega$	orange	$T(1)$ with $2-\omega$, $3-\omega$ (yellow)
12	$3+\omega$	orange	$T(1)$ with 3, 4 (yellow)
13	7	orange	$T(1,3,4)$ with 3, 4 (yellow)
14	$4+\omega$	orange	$T(1)$ with 4, 5 (yellow)
15	$5-\omega$	orange	$T(1)$ with 4, 5 (yellow)
16	$-5-\omega$	yellow	$T(1)$ with -6, -5 (orange)
17	$-3+2\omega$	yellow	$T(1)$ with $-3+\omega$, $-2+\omega$ (orange)
18	$-1-2\omega$	yellow	$T(1)$ with $-2-\omega$, $-1-\omega$ (orange)
19	2ω	orange	$T(1)$ with ω, $1+\omega$ (yellow)
20	$3+2\omega$	yellow	$T(1,3,4)$ with $3-2\omega$, $3+\omega$ (orange)

Case: q yellow.

	point	colour	because
1	$q(-3+4\omega)$	orange	$T(1,3,4)$ with $-3+2\omega$, q (yellow)
2	$q(1+3\omega)$	orange	$T(3)$ with $-3+2\omega$, q (yellow)
3	$q(2-\omega)$	orange	$T(1,3,4)$ with $-3+2\omega$, q (yellow)
4	$q(\omega)$	yellow	$T(3)$ with $q(-3+4\omega)$, $-2-\omega$ (orange)
5	$q(5-\omega)$	yellow	$T(3)$ with $q(2-\omega)$, $5-\omega$ (orange)
6	$q(4-3\omega)$	orange	$T(1,3,4)$ with $q(\omega)$, q (yellow)
7	$q(5-3\omega)$	yellow	$T(1)$ with $q(4-3\omega)$, $5-\omega$ (orange)

Subcase: $q(4-\omega)$ yellow.

	point	colour	because
1	$q(1-\omega)$	orange	$T(1,3,4)$ with $q(4-\omega)$, $q(5-\omega)$ (yellow)
2	$q(-3+3\omega)$	yellow	$T(4)$ with $q(1+3\omega)$, $q(1-\omega)$ (orange)
3	$q(-2-\omega)$	yellow	$T(1,3,4)$ with $q(1-\omega)$, $q(2-\omega)$ (orange)
4	$q(-\omega)$	yellow	$T(1)$ with 0, $q(1-\omega)$ (orange)
5	$q(2-2\omega)$	yellow	$T(1)$ with $q(1-\omega)$, $q(2-\omega)$ (orange)
6	$q(-6+3\omega)$	orange	$T(1,3,4)$ with $q(-3+3\omega)$, $-3+2\omega$ (yellow)
7	$2+\omega$	orange	$T(1,3,4)$ with $q(-\omega)$, $q(4-\omega)$ (yellow)
8	$-1+\omega$	yellow	$T(1,3,4)$ with $-2+\omega$, $2+\omega$ (orange)
9	$q(-1-\omega)$	yellow	$T(1,3,4)$ with $q(2-\omega)$, $2+\omega$ (orange)
10	$q(3-2\omega)$	yellow	$T(1)$ with $q(2-\omega)$, $2+\omega$ (orange)
11	$-4+\omega$	orange	$T(3)$ with $-1-2\omega$, $-1+\omega$ (yellow)
12	$1-3\omega$	orange	$T(1)$ with $q(-2-\omega)$, $q(-1-\omega)$ (yellow)
13	$4+2\omega$	orange	$T(1,3,4)$ with $q(2-2\omega)$, $q(3-2\omega)$ (yellow)
14	$-7+\omega$	yellow	$T(1,3,4)$ with $-4+\omega$, $-3+\omega$ (orange)
15	$-3-3\omega$	yellow	$T(4)$ with $-3+\omega$, $1-3\omega$ (orange)
16	$1+2\omega$	yellow	$T(1,3,4)$ with 2ω, $4+2\omega$ (orange)
17	$-4-2\omega$	orange	$T(1,3,4)$ with $-7+\omega$, $-3-3\omega$ (yellow)
18	$-2+2\omega$	orange	$T(1,3,4)$ with $-3+2\omega$, $1+2\omega$ (yellow)
19	$q(-6+6\omega)$	yellow	$T(1,3,4)$ with $q(-6+3\omega)$, $-4-2\omega$ (orange)
20	$-5+2\omega$	yellow	$T(3)$ with $-2-\omega$, $-2+2\omega$ (orange)
21	$-8+3\omega$	orange	$T(1,3,4)$ with $q(-6+6\omega)$, $q(-3+3\omega)$ (yellow)
22	$-5+3\omega$	orange	$T(1,3,4)$ with $-5-\omega$, $-5+2\omega$ (yellow)

Contradiction: $-8+3\omega$, -5, $-5+3\omega$ are all orange and form $T(3)$.

Subcase: $q(4-\omega)$ orange.

	point	colour	because
1	$q(5-2\omega)$	yellow	$T(1,3,4)$ with $q(4-\omega)$, 7 (orange)
2	$4+2\omega$	orange	$T(1)$ with $q(5-\omega)$, $q(5-2\omega)$ (yellow)
3	$q(4-2\omega)$	orange	$T(1)$ with $q(5-2\omega)$, $q(5-3\omega)$ (yellow)
4	$1+2\omega$	yellow	$T(1,3,4)$ with 2ω, $4+2\omega$ (orange)
5	$2+\omega$	yellow	$T(1)$ with $q(4-\omega)$, $q(4-2\omega)$ (orange)

Contradiction: $1+\omega$, $1+2\omega$, $2+\omega$ are all yellow and form $T(1)$.

Case: q orange.

	point	colour	because
1	$q(\omega)$	yellow	$T(1)$ with 0, q (orange)
2	$q(1-\omega)$	yellow	$T(1)$ with 0, q (orange)
3	$q(2-\omega)$	yellow	$T(1,3,4)$ with q, $5-\omega$ (orange)
4	$q(4-3\omega)$	yellow	$T(1,3,4)$ with q, $5-\omega$ (orange)
5	$1+4\omega$	orange	$T(4)$ with $q(\omega)$, $q(4-3\omega)$ (yellow)
6	$q(2-2\omega)$	orange	$T(1)$ with $q(1-\omega)$, $q(2-\omega)$ (yellow)
7	$q(-1+2\omega)$	orange	$T(1,3,4)$ with $-3+2\omega$, $q(2-\omega)$ (yellow)
8	$q(2+3\omega)$	orange	$T(4)$ with $-3+2\omega$, $q(2-\omega)$ (yellow)
9	$-3+4\omega$	yellow	$T(4)$ with 1, $1+4\omega$ (orange)
10	5ω	yellow	$T(1,3,4)$ with $1+4\omega$, $4+\omega$ (orange)
11	$q(1-2\omega)$	yellow	$T(1)$ with $q(2-2\omega)$, $3-2\omega$ (orange)
12	$-4+5\omega$	orange	$T(1,3,4)$ with $-3+4\omega$, ω (yellow)
13	$-1+6\omega$	orange	$T(1,3,4)$ with 5ω, $3+2\omega$ (yellow)
14	$-1+3\omega$	orange	$T(1,3,4)$ with $q(1-\omega)$, $q(1-2\omega)$ (yellow)
15	$q(-\omega)$	orange	$T(1)$ with $q(1-\omega)$, $q(1-2\omega)$ (yellow)
16	$q(-4+5\omega)$	yellow	$T(3)$ with $-4+5\omega$, $q(-1+2\omega)$ (orange)
17	$-1+7\omega$	yellow	$T(1,3,4)$ with $-1+3\omega$, $-1+6\omega$ (orange)
18	$q(3\omega)$	yellow	$T(1,3,4)$ with 0, $q(-\omega)$ (orange)
19	$q(-3+4\omega)$	orange	$T(1,3,4)$ with $q(-4+5\omega)$, $q(\omega)$ (yellow)
20	$q(5\omega)$	orange	$T(4)$ with $q(-4+5\omega)$, $q(\omega)$ (yellow)
21	$q(1+4\omega)$	yellow	$T(4)$ with $q(-3+4\omega)$, q (orange)
22	$q(4\omega)$	yellow	$T(1)$ with $-4+5\omega$, $q(5\omega)$ (orange)
23	$q(1+3\omega)$	orange	$T(1)$ with $q(4\omega)$, $q(1+4\omega)$ (yellow)
24	$-5+8\omega$	orange	$T(1,3,4)$ with $q(4\omega)$, $q(3\omega)$ (yellow)
25	7ω	yellow	$T(1,3,4)$ with $q(1+3\omega)$, $q(2+3\omega)$ (orange)
26	$-4+7\omega$	yellow	$T(1,3,4)$ with $-5+8\omega$, $-1+4\omega$ (orange)

Contradiction: $-4+7\omega$, $-1+7\omega$, 7ω are all yellow and form $T(1,3,4)$.

4. Rows

Let L be a unit triangular lattice with long direction u, and put $v = u\omega$; the three directions of L are u, v and $v-u = u\omega^2$. The *rows* of L are its lines in direction u; the row after R is $R+v$.

Lemma 4.1 (four apart)**.** *For every $z \in L$ we have $\chi(z+4u) \neq \chi(z)$ and $\chi(z+8u) = \chi(z)$.*

Proof. By Theorem 2.2 and Rule A every maximal monochromatic run in direction u has exactly four points, so each row is coloured R R R R B B B B, repeating. □

Lemma 4.2 (consecutive rows)**.** *Let R be a row of L. Then either $\chi(z+v) \neq \chi(z)$ for every $z \in R$, or $\chi(z+v-u) \neq \chi(z)$ for every $z \in R$. We say that the pair of rows $(R, R+v)$ has* type *v or $v-u$ accordingly.*

Proof. This is the neighbourhood argument of [3, Theorem 3.12]. Let $t, t+u, t+2u, t+3u$ be a orange run in R. Each of $t+v$, $t+u+v$, $t+2u+v$ forms a unit triangle with two consecutive points of the run, so all three are yellow by (F). By Lemma 4.1 the yellow run of $R+v$ containing them has exactly four points: it is either $t+v, \dots, t+3u+v$ or $t-u+v, \dots, t+2u+v$. In the first case $\chi(z+v) \neq \chi(z)$ for $z = t, \dots, t+3u$, in the second $\chi(z+v-u) \neq \chi(z)$. Since Lemma 4.1 flips the colours of both rows under $z \mapsto z+4u$, the same relation holds for every $z \in R$. □

5. Rule B

Fix a row R. If $z, z+u \in R$ have the same colour then, by (F), the point $z+v$ above them and the point $z+u-v$ below them both have the other colour. The remaining points of $R \pm v$, above

or below a pair $z, z+u$ with $\chi(z) \neq \chi(z+u)$, are *open*. They come in vertically aligned *open pairs* $\{z+v,\ z+u-v\}$ (Figure 7).



FIGURE 7. One row of the long direction, and the points it forces by unit triangles in the rows above and below. The open points (ringed) come in vertically aligned pairs.

Lemma 5.1. *An open pair $\{z+v,\ z+u-v\}$ at the row R is monochromatic exactly when the pairs of rows $(R-v, R)$ and $(R, R+v)$ have different types.*

Proof. Let $c = \chi(z)$, so $\chi(z+u) \neq c$. If $(R, R+v)$ has type v then $\chi(z+v) \neq c$; if it has type $v-u$ then $\chi(z+v) \neq \chi(z+u)$, so $\chi(z+v) = c$. Let $y = z+u-v \in R-v$. If $(R-v, R)$ has type v then $\chi(y) \neq \chi(y+v) = \chi(z+u)$, so $\chi(y) = c$; if it has type $v-u$ then $\chi(y) \neq \chi(y+v-u) = \chi(z)$, so $\chi(y) \neq c$. Hence $\chi(z+v) = \chi(y)$ exactly when the two types differ. □

Theorem 5.2 (Rule B)**.** *In an admissible colouring, all pairs of consecutive rows of a unit triangular lattice have the same type. Equivalently, every open pair has two colours.*

Proof. The two statements are equivalent by Lemma 5.1; suppose some open pair is monochromatic. Translating along the row, exchanging the colours and reflecting in a line perpendicular to the rows if necessary, we may assume that $L = \Lambda$, $u = 1$, that the row $R = \mathbb{Z} + \omega$ is orange exactly at the points $a + \omega$ with $a \equiv 4, 5, 6, 7$, and that the open pair $\{-1+2\omega,\ 0\}$ is orange. By (F) and Lemma 4.1 the neighbouring rows are then determined: $\mathbb{Z}$ is orange exactly at $a \equiv 0, 1, 2, 3$, and $\mathbb{Z} + 2\omega$ is orange exactly at $a + 2\omega$ with $a \equiv 7, 0, 1, 2$ (all congruences modulo 8; Figure 8). We use only these three rows, rule (F), and Lemma 4.1 applied to $q\Lambda$.

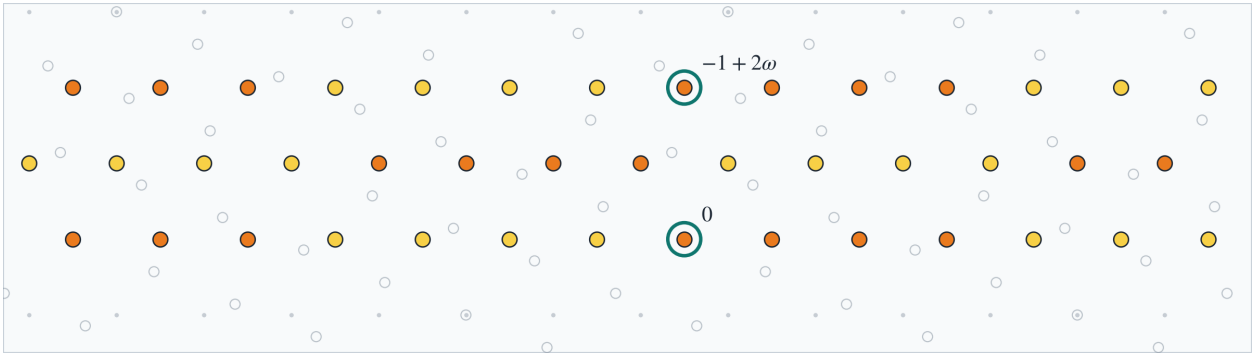


FIGURE 8. A monochromatic open pair: 0 and $-1+2\omega$ are both orange, and the three rows are then determined.

Common deductions. Inside Λ alone, (F) gives five further colours (Table 1, Figure 9).

TABLE 1. Deductions inside Λ, valid in every case.

	point	colour	because
1	$-2-\omega$	orange	$T(1)$ with -3, -2 (yellow)
2	$-1+3\omega$	yellow	$T(1)$ with $-1+2\omega$, 2ω (orange)
3	$2-\omega$	yellow	$T(1)$ with 1, 2 (orange)
4	$3-\omega$	yellow	$T(1)$ with 2, 3 (orange)
5	$3-2\omega$	orange	$T(1)$ with $2-\omega$, $3-\omega$ (yellow)

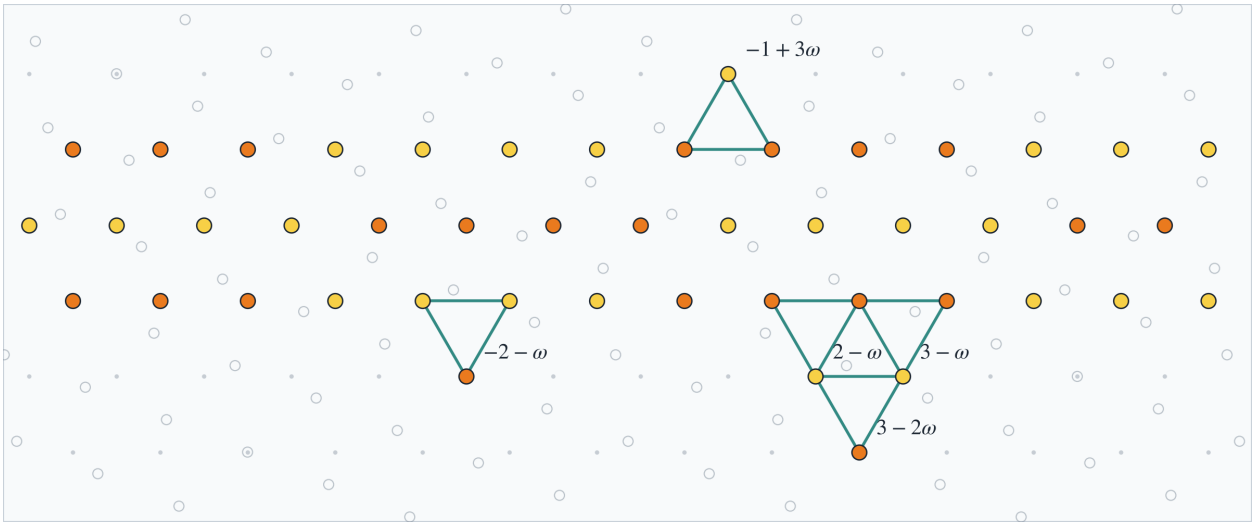


FIGURE 9. The five deductions of Table 1, inside Λ alone.

The cases. By Theorem 2.2 the long direction of $q\Lambda$ is $q\omega^j$ for some $j \in \{0,1,2\}$, and by Lemma 4.1 points of $q\Lambda$ four steps apart along it have opposite colours. In each case a few more deductions end in a monochromatic forbidden triangle:
Case: the long direction of $q\Lambda$ is q.

	point	colour	because
1	$q(2-2\omega)$	orange	four apart from $4+2\omega$ along q
2	$q(1+\omega)$	yellow	four apart from $-2-\omega$ along q
3	$q(2+\omega)$	orange	$T(1)$ with $-1+3\omega$, $q(1+\omega)$ (yellow)

Contradiction: $q(2+\omega)$, $q(2-2\omega)$, $3-2\omega$ are all orange and form $T(1,3,4)$.

Case: the long direction of $q\Lambda$ is $q\omega$.

	point	colour	because
1	$q(1-2\omega)$	orange	four apart from $-1+3\omega$ along $q\omega$
2	$q(2+\omega)$	yellow	four apart from $3-2\omega$ along $q\omega$
3	$q(-2+\omega)$	yellow	$T(1,3,4)$ with $q(1-2\omega)$, $3-2\omega$ (orange)
4	$-q$	orange	$T(1,3,4)$ with $-5+\omega$, $q(-2+\omega)$ (yellow)
5	$q(-1+\omega)$	yellow	$T(1)$ with $-q$, 0 (orange)

Contradiction: $q(-2+\omega)$, $q(-1+\omega)$, $q(2+\omega)$ are all yellow and form $T(1,3,4)$.

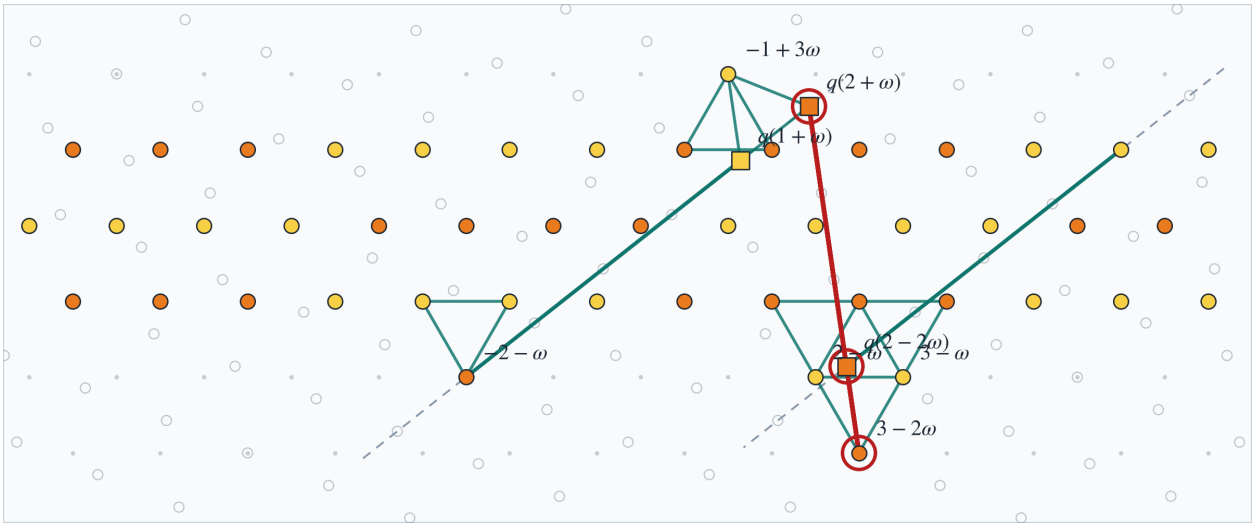


FIGURE 10. Rule B, long direction q: every deduction of this case (teal), the four-apart steps along the dashed lines, and the contradiction (red).

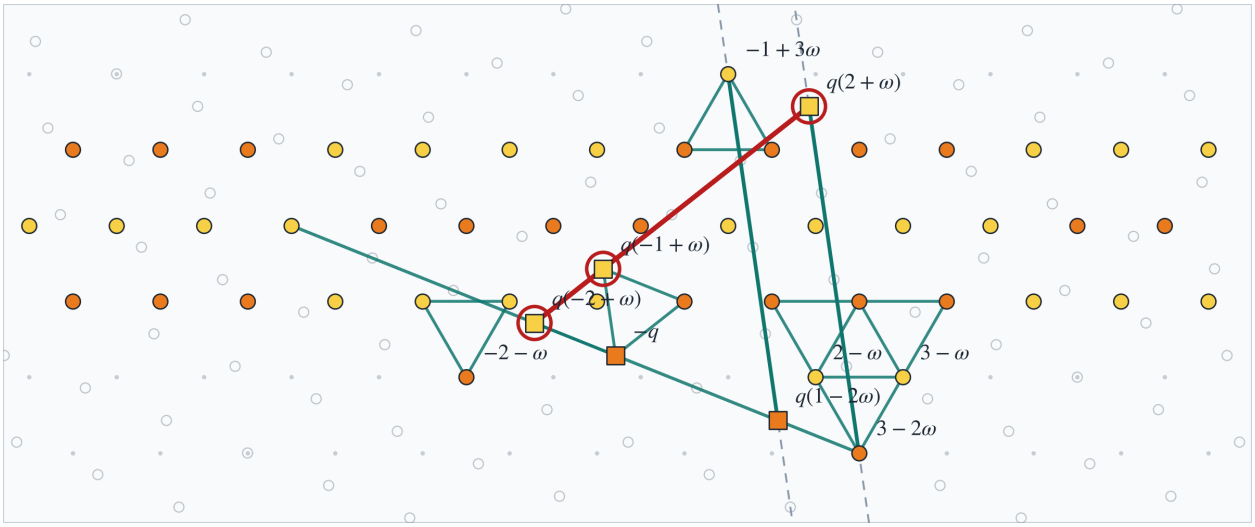


FIGURE 11. Rule B, long direction $q\omega$: every deduction of this case (teal), the four-apart steps along the dashed lines, and the contradiction (red).

Case: the long direction of $q\Lambda$ is $q\omega^2$.

	point	colour	because
1	$q(-4+\omega)$	yellow	four apart from -7 along $q\omega^2$
2	$-q$	orange	four apart from $-5+\omega$ along $q\omega^2$
3	$q(-4+4\omega)$	yellow	four apart from 0 along $q\omega^2$
4	$q(-1+\omega)$	yellow	$T(1)$ with $-q$, 0 (orange)

Contradiction: $q(-4+4\omega)$, $q(-4+\omega)$, $q(-1+\omega)$ are all yellow and form $T(3)$.

All three directions are impossible, so no open pair is monochromatic. □

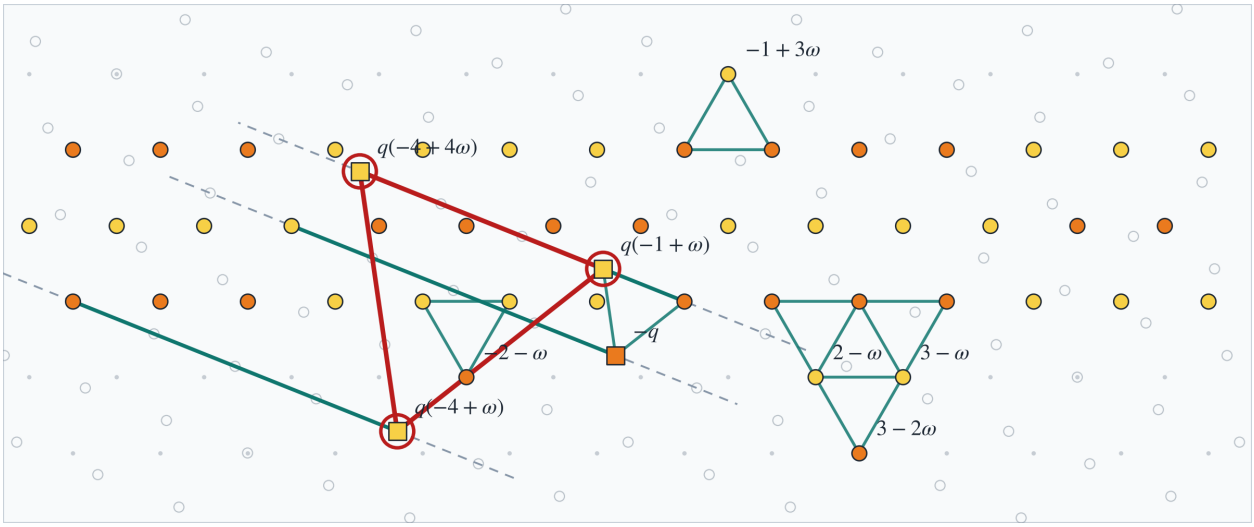


FIGURE 12. Rule B, long direction $q\omega^2$: every deduction of this case (teal), the four-apart steps along the dashed lines, and the contradiction (red).

6. Distance eight and the proof of Theorem 1.1

Proposition 6.1. *In an admissible colouring, any two points at distance* 8 *have the same colour.*

Proof. Let $|p-p'| = 8$ and take a unit triangular lattice $L \ni p$ in which $p'-p = 8w$ for a direction w of L. By Rule B all pairs of consecutive rows of L have the same type $t \in \{v, v-u\}$, so by Lemma 4.2 $\chi(z+t) \neq \chi(z)$ for every $z \in L$, and hence $\chi(z+8t) = \chi(z)$. By Lemma 4.1, $\chi(z+8u) = \chi(z)$. The third direction is $t-u$ or $t+u$, and $\chi(z+8t \pm 8u) = \chi(z)$ as well. So $\chi(p') = \chi(p)$ whichever direction w is. □

Proof of Theorem 1.1. Suppose χ is admissible and $|x-y| = 1$. The circles of radius 8 about x and y meet in a point r, and Proposition 6.1 gives $\chi(x) = \chi(r) = \chi(y)$. So every unit equilateral triangle is monochromatic, contradicting Lemma 2.1. □

7. Verification

The finite parts of Rule A and Rule B are recorded in certificates with standard-library verifiers. For Rule A the verifier checks that every point lies in $\Lambda \cup q\Lambda$, re-derives all forbidden triangles from coordinates, checks the symmetry statements and that the four cases' orbits cover all nine boundary colourings, and replays each case with rule (F). For Rule B it checks Lemma 5.1 for all four combinations of types and every deduction of Table 1 and of the three cases, each against exactly one rule. The tables in this paper are generated from the certificates. Interactive walkthroughs of both rules accompany the code.

Use of AI

The exploration purely on the triangular grid, including the neighbourhood argument, was done during the work on my thesis [3]. While working on it, it came to my mind to overlap triangular lattices as a potential direction, which the thesis mentions in its conclusion: I was driven by the intuition that three or four directions are far too limiting, that we need to overlap triangular lattices, and that the constraints would then accumulate quickly enough to force a contradiction. GPT-6 Astra discovered that after the overlap the colouring is 8-periodic, and found Rule A and Rule B as a way to split the proof of 8-periodicity; it also calculated the coordinates of the points needed and derived the proofs. Claude Opus 5.5 was used for the visualizations and for language polishing.

Acknowledgements

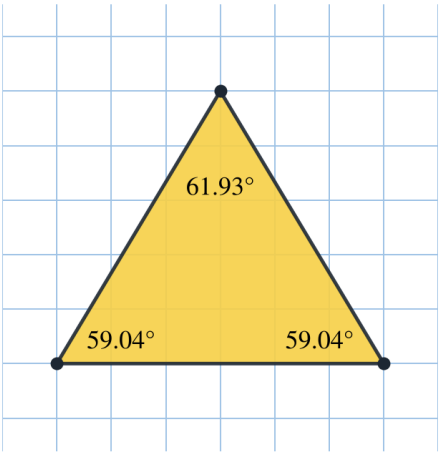
I am grateful to my supervisor, doc. Mgr. Jan Kynčl, Ph.D., whose suggestion to look at this problem created the spark for this work, and for his guidance during my thesis work, which made this paper possible.

Dedication

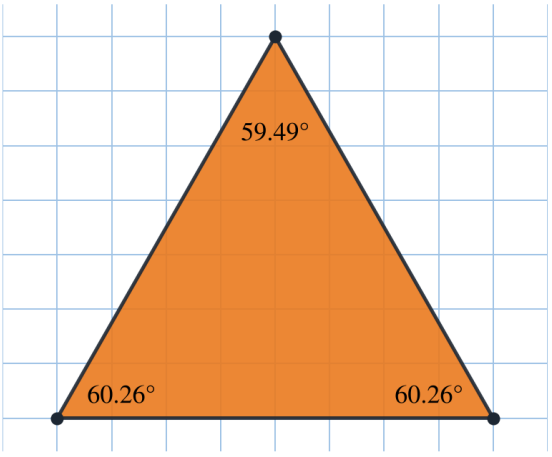
This work is dedicated to the memory of the extraordinary Kharkiv teacher, computer scientist, mathematician, and wartime volunteer Yulia Zdanovska[1] (*04.05.2000–†03.03.2022), who opened my eyes to the wonder of combinatorics, graphs, and algorithms. She spent countless hours patiently teaching me to solve problems in these fields when we were preparing for mathematics and informatics olympiads together, back in the carefree days of our childhood.

I remember us reading a paper on the chromatic number of space while sitting on the Giant Bench in Holosiiv Park, two and a half metres above the ground. It was a sunny, calm day in the forest by the lake full of ducks, and we were eating improvised sandwiches. That is when we discovered that we drew equilateral triangles on grid paper in different ways: I used isosceles triangles with base 6 and height 5, she used isosceles triangles with base 8 and height 7. We made a bet on a chocolate bar over whose approximation was better, and calculated the angles. Mine came out as (59.04°, 59.04°, 61.93°), hers as (60.26°, 60.26°, 59.49°). She won the chocolate bar.

I wish there were more such afternoons, and more bets for her to win.



base 6, height 5



base 8, height 7

I am eternally grateful for her inspiration and feel honored to have known her so closely.

[1]For more about her life and legacy, see `https://teachforukraine.org/en/t4u-news/ne-hochu-v-garvard-hochu-buty-ministerkoyu-osvity-istoriya-zagybloyi-matematykyni-yuliyi-zdanovskoyi/`.

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[2] G. Currier, K. Moore and C. H. Yip, *Any two-coloring of the plane contains monochromatic 3-term arithmetic progressions*, Combinatorica **44** (2024), no. 6, 1367–1380.
[3] A. Yan, *Monochromatic configurations in two-colored plane*, Bachelor's thesis, Charles University, Prague, 2025.